

RIYA GUTTIGOLI

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EDUCATION

Cornell University, College of Engineering

Ithaca, NY

Master of Engineering in Systems Engineering

Expected Dec. 2026

Courses: MBSE, Computational Optimization, Entrepreneurship for Scientists & Engineers, Proj. Management for Complex Systems

Bachelor of Science in Mechanical Engineering | Minor: Business

May 2026

Courses: Advanced Product Design, Electric Vehicles, Future Energy Systems, Mechatronics, System Dynamics, Fluid Mechanics

Affiliations: Architectural Robotics Lab, Society of Women Engineers, Cornell Nazaqat (dance), Kappa Alpha Theta, CU Hyperloop

RELEVANT EXPERIENCE

M.Eng Researcher

Jan 2026 - Present

CORNELL ORGANIC ROBOTICS LAB, Ithaca, NY

- Engineer and prototype a 3-DOF electromechanical test platform in Onshape to characterize volumetrically manufactured fiber-optic sensors. Develop a spherical actuator for smooth 2-DOF motion with a separately actuated rotational axis.
- Iterate servo-driven actuation, embedded angle sensing, custom 3D-printed fixtures, and modular PCB interfaces to enable precise, repeatable multi-axis sensor characterization.

Mechanical Engineering Intern - Electromechanical Systems & System Packaging

May 2025 - Aug 2025

DRAPER, Cambridge, MA

- Designed and qualified electromechanical test hardware for vibration environments up to 500 G SRS, applying first-principles calculations, FEA, and GD&T to optimize structural performance, rigidity, and hardware integration.
- Built and integrated an air-driven test assembly for guided munitions, collaborating across mechanical and electrical teams and using LabVIEW and MATLAB for system testing and analysis.
- Reverse-engineered a precision thermal socket for a rad-hardened IMU with no existing design documentation; produced GD&T drawings and assembly instructions to enable production handoff.

Mechanical Engineering Intern - Mechanical & Reliability Systems Team

May 2024 - Aug 2024

THE MITRE CORPORATION, Bedford, MA

- Modeled and produced 40+ scaled 3D-printed prototypes of HEMP-protected enclosures for UHF radio systems aboard C-130 aircraft, using SolidWorks, Creo, SpaceClaim, and Ansys Icepak for CAD, rapid prototyping, and thermal analysis.
- Analyzed aircraft certification, aerial refueling, and digital-twin integration challenges to author two contextual white papers.

Mechanical Team Lead

Jan 2023 - May 2026

CORNELL HYPERLOOP, Ithaca, NY

- Led a multidisciplinary team of 40+ engineers in design, fabrication, testing, and integration of a scaled magnetic-levitation pod across propulsion, levitation, structures, braking, and electrical subsystems.
- Initiated the “MiniPod” platform to demonstrate small-scale maglev; advanced from transformer modules to custom electromagnets, characterized magnet saturation and forces up to 10 lbf/magnet, and integrated the levitation controls.
- Directed linear induction motor thrust characterization across operating conditions, identifying a ~15 lbf thrust plateau that drove the transition from a steel to lightweight aluminum chassis and exploration of a custom propulsion system.

PROJECTS

Wind Turbine Blade Optimization | MATLAB, BEM, CFD, Wind Tunnel Testing

Oct 2025 - Dec 2025

- Optimized a low-Reynolds-number wind turbine blade using Blade Element Momentum theory and aerodynamic analysis, validating predicted performance through CFD and wind tunnel testing.

Autonomous Cube Collection Robot | CAD, Rapid Prototyping, Arduino, Sensor Integration

Jan 2025 - May 2025

- Led mechanical design and subsystem integration for an autonomous competition robot, iterating the chassis and collection system through CAD, rapid prototyping, sensor integration, and physical testing.

Electromagnet Cooling System | MATLAB, Thermal Analysis, Fabrication, Experimental Testing

Aug 2023 - May 2024

- Designed and experimentally validated an active electromagnet cooling system using aluminum components and forced air, reducing operating temperature by 30°C through iterative prototyping, thermistor testing, and MATLAB thermal analysis.

SKILLS

CAD: SolidWorks, Creo, Onshape, Fusion 360, AutoCAD

Design: GD&T, Tolerance Analysis, BOMs, Design Reviews, Manufacturing Drawings

Analysis: ANSYS Mechanical, Fluent, Icepak, MATLAB, FEA, Instrumentation, Data Acquisition, LabVIEW, Environmental Testing

Fabrication: Machining, Lathe, Mill, Laser Cutting, 3D Printing, Mechanical Assembly

Programming: Python, C++, Arduino

Leadership: Cornell Nazaqat Captain, Hyperloop Team Lead, Future Energy Systems Teaching Assistant